WW-C-563A
December 4, 1973
SUPERSEDING
Fed. Spec. WW-C-563
April 2, 1963

FEDERAL SPECIFICATION

CONDUIT, METAL, RIGID: ELECTRICAL, THIN-WALL STEEL TYPE (ELECTRICAL METALLIC TUBING); STRAIGHT LENGTHS, ELBOWS, AND BENDS

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rigid steel thin-wall conduit in straight lengths, elbows, or bends for use as a raceway for the wires or cables of an electrical system.

1.2 Classification. Conduit shall be of the following types as specified (see 6.2):

Type I - Straight lengths
Type II - Elbows
Type III - Bends

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Standards

FED. STD. No. 123 - Marking for Domestic Shipment (Civil Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications

MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.

Military Standards

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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Underwriters' Laboratories, Inc., (UL) Standard:

UL 797 - Standard for Electrical Metallic Tubing.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingston Road, Box 247, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, Long Island, NY 11746 or 1655 Scott Lane, Santa Clara, CA 95050.)

American National Standards Institute (ANSI) Standard:

C80.3 - Electrical Metallic Tubing Zinc Coated, Specification for.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Description. Conduit shall be the rigid thin-wall steel type electrical metallic tubing (EMT) and shall conform to ANSI Standard C80.3, and to Underwriters' Laboratory Standard UL 797. Conduit shall be furnished in the size specified (see figure 1 and 6.2) and in straight lengths, bends, or elbows as specified (see 1.2).

3.1.1 Fire and casualty hazards. The UL seal or label attached to the conduit shall be accepted as evidence that the conduit conforms to the requirements of this specification. Subject to approval of the contracting officer, the supplier may submit proof from another nationally recognized agency that the conduit conforms to the applicable requirements of that agency. Agency approval shall not absolve the supplier from compliance with the requirements of this specification.

3.2 Identification marking. Conduit shall be marked in accordance with UL 797.

3.3 Workmanship. Conduit shall be free from defects, such as imperfect zinc or enamel coatings, hard scale, burrs, fins, or any other defect that could impair the operation or serviceability of the conduit.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level II for examination and inspection level S-4 for testing.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall be 1.5 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 1.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. Conduit shall be examined as specified herein for the following defects:

101. Size not as specified.
102. Identification marking missing.
103. UL seal or label (or equivalent when approved) missing.
104. Workmanship not as specified.

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4.4.2 Tests.

4.4.2.1 Composition. Test the conduit for bending properties, ductility of zinc coating, thickness of zinc coating, and quality of enamel coating in accordance with ANS C80.3. In determining the thickness of the zinc coating the Preece test specified in UL 797 shall be used. Nonconformance to 3.1 shall constitute failure of this test.

4.5 Inspection of preparation for delivery. The packaging and packing shall be examined and tested in accordance with MIL-E-17555. The marking shall be examined to determine if any marking is missing, illegible, incorrect, or incomplete in accordance with the referenced document. Any defect in the packaging, packing or marking shall be cause for rejection.

5. PREPARATION FOR DELIVERY

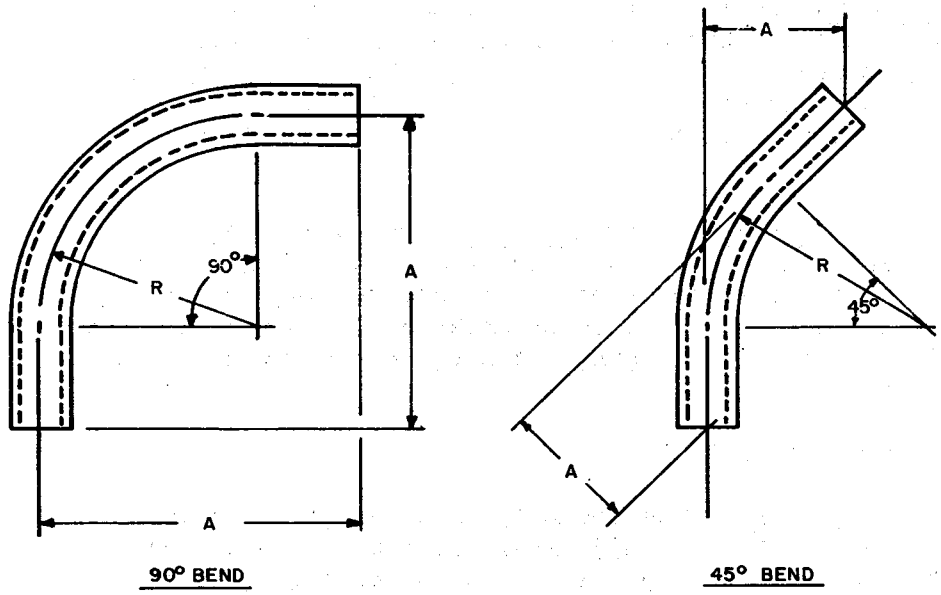
5.1 Packaging and packing. The conduit, shall be packaged and packed in accordance with MIL-E-17555 except fiberboard boxes are not acceptable for level A packing. The level of packaging shall be level A or C and the level of packing shall be level A, B, or C as specified (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order shipping containers shall be marked in accordance with FED. STD. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order shipping containers shall be marked in accordance with MIL-STD-129.

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NOM SIZE	ANGLE ±2°	DIAMETERS		RADIUS R −0 +.25	OFFSET A ±.25	WALL THK
		EXT	INT (REF)			
.50	90°	.706	.622	4.00	6.63	.040
.50	45°	.706	.622	4.00	3.06	.040
.75	90°	.922	.824	4.50	7.25	.046
.75	45°	.922	.824	4.50	3.31	.046
1.00	90°	1.163	1.049	5.75	8.63	.054
1.00	45°	1.163	1.049	5.75	3.75	.054
1.25	90°	1.510	1.380	7.25	10.25	.061
1.25	45°	1.510	1.380	7.25	4.25	.061
1.50	90°	1.740	1.610	8.25	11.38	.061
1.50	45°	1.740	1.610	8.25	4.63	.061
2.00	90°	2.197	2.067	9.50	13.50	.061
2.00	45°	2.197	2.067	9.50	5.63	.061

OUTSIDE DIAMETER: ±.005 BEFORE BENDING
WALL THICKNESS: + 18 PERCENT

FIGURE 1. CONDUIT

X-2643

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6. NOTES

6.1 Intended use. Thin-wall conduit, is not threaded and can be used only with special fittings. Such fittings are not covered by this specification.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and inclue the following information in procurement documents.

(a) Title, symbol, and date of this specification.
(b) Type of conduit required (see 1.2).
(c) Size of conduit required (see 3.1).

MILITARY INTEREST:

Custodians:

Army - ME
Air Force - 80

User activity:

Army - EL

Preparing activity:

Army - ME

Civil Agencies Coordinating Activity:

USDA - AFS
DC GOVT - DCG

Project No. 5975-0412

7

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price cents each.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION WW-C-563A Conduit, Metal, Rigid: Electrical, Thin-Wall Steel Type (Electrical, Metallic Tubing); Straight Lengths, Elbows and Bends

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426
11 Oct 72 (Rev)

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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